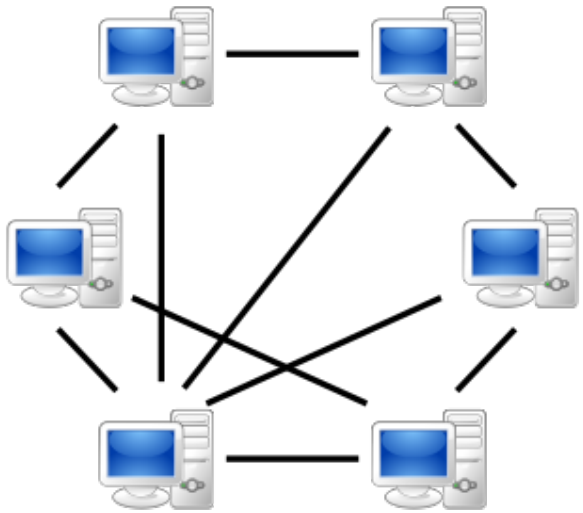
Attaining Equilibria Using Control Sets

COCOA'24 - best paper award; TCS

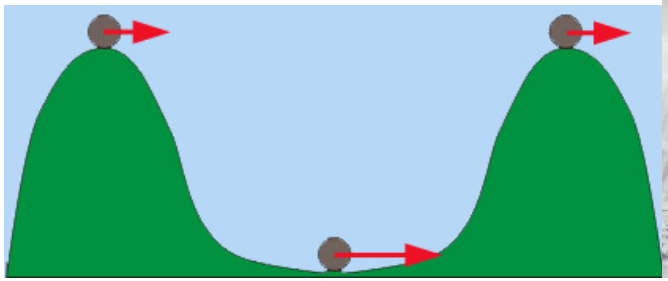
Gleb Polevoy Jonas Schweichhart



Aims

In an interaction (society, service providing, politics, etc.), while playing a profile, consider some equilibrium we like.

- Find the **easiest** way to **motivate** playing this equilibrium
- Do this **quickly**





Goals and Plan

Motivating equilibria is often external - by subsidies, taxes, mediation, etc.

Internal

Assume that some agents are motivated/bribed/controlled to act and thereby motivate the others to act as desired.



The plan:

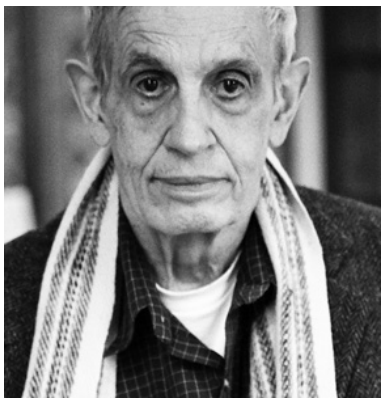
1. Model
2. Control vs strength
3. Hardness and some algorithms
4. Spanning all finite games
    1. Singleton congestion games
    2. Harmonic games
5. Variations of the model

Model

1. $G = (N, S = S_1 \times S_2 \times \ldots \times S_n, (u_i)_{i=1,\ldots,n})$,
2. The solutions, forming the solution set, are a set of strategy profiles $D \subseteq S$,
3. Our default is Nash equilibrium, though our definitions are general

$$\forall i \in N, \forall s_i' \in S_i : u_i(s) \geq u_i(s_i', s_{-i}), \tag{1}$$

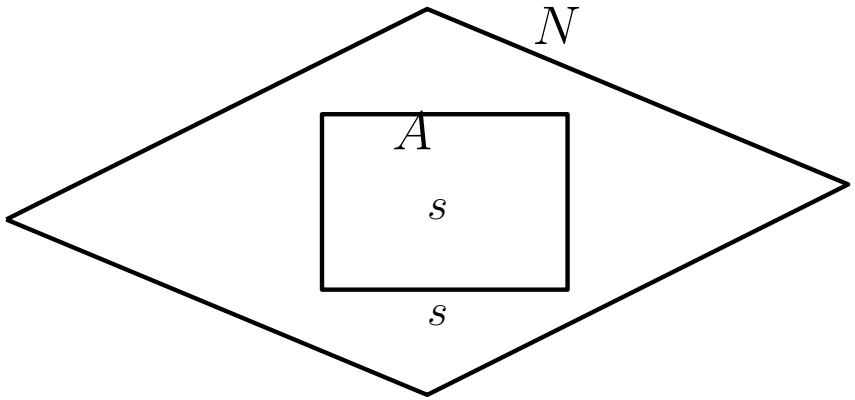
where $s_{-i} \stackrel{\Delta}{=} (s_1, \ldots, s_{i-1}, s_{i+1}, s_n)$.



Model – Direct Control Set

Definition (Direct Control Set)

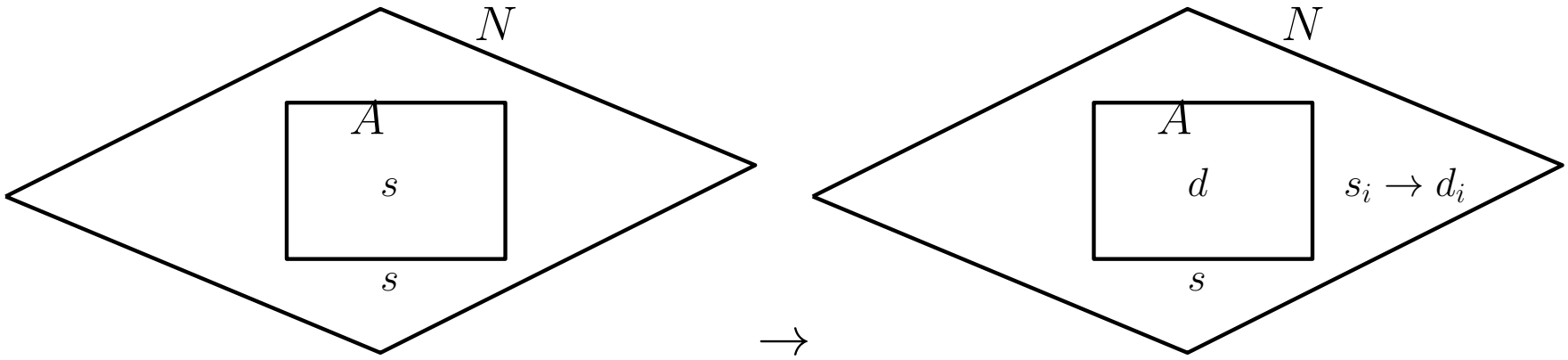
Consider any profile s and a Nash eq. $d \in NE$. A set of agents $A \subseteq N$ can bring s to d directly (by playing d), while all the others play s, if for each agent i outside A, d_i is a BR to $(s_{N\setminus(\{i\}\cup A)}, d_A)$.
A is called a direct control set with respect to G, s, and d.
The minimum size of a direct control set is called the direct control number, denoted by $\mathrm{DirConNum}(G, s, d)$.



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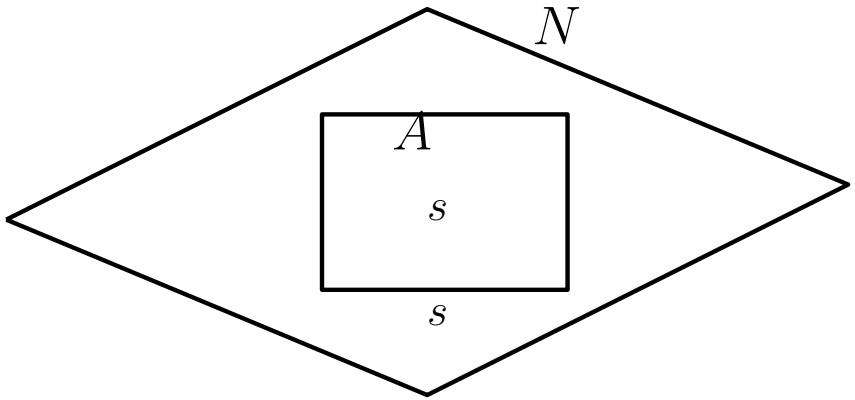
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Model – Direct Control Set Problem

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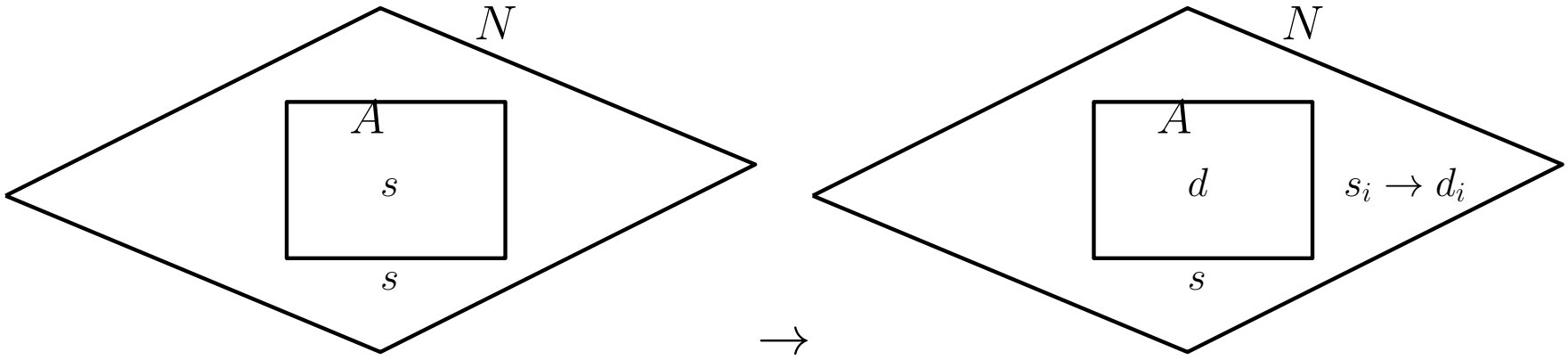
The direct control set problem receives a game $G = (N, S, (u_i)_{i=1,\ldots,n})$, *profile* s *and equilibrium* d, *and weights* $w\colon N \to \mathbb{R}_+$.
A solution is a set of agents $A \subseteq N$.
A feasible solution is a solution that can bring s *to* d *directly,*
Find a feasible solution A *with the minimum* $w(A) \stackrel{\Delta}{=} \sum_{a \in A} w(a)$.



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Control vs Strong NE

Theorem

Assume that game G satisfies both following conditions for some $d \in NE$:

1. *Given any profile s, any subset of at least m agents is a direct control set with respect to G, s and d.*
2. *For any player j, playing d_j in d is at least as profitable for her as in any other profile.*

Then, d is an $n - m$-strong NE, and this is tight for every $m < n$.

The converse does **not** hold!

Hardness

Theorem

The decision version of direct control set is NP-complete.

Theorem

Is not approximable within factor $n^{1-\epsilon}$, $\forall \epsilon > 0$, unless $P = NP$.
Holds even when bringing an NE to an NE, and each player has 2 strategies.



Coping with Hardness - DP on Trees, Etc.

We can provide a DP solution for the case of the game graph being a tree.

Coping with Hardness - DP on Trees, Etc.

We can provide a DP solution for the case of the game graph being a tree. Given a game, transform it to the graph where

1. The nodes are the players,
2. Two nodes are connected $\iff$ there is some influence

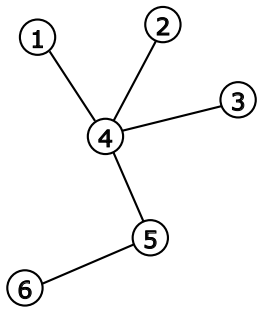
Coping with Hardness - DP on Trees, Etc.

We can provide a DP solution for the case of the game graph being a tree. Given a game, transform it to the graph where

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Now,

1. If the graph is a tree, then Dynamic Programming solves optimally.
2. If we can obtain such a graph by taking out $\leq c\mathrm{OPT}$ weight from the game, then we obtain a $1 + c$-approximation, assuming monotonicity.



Coping with Hardness

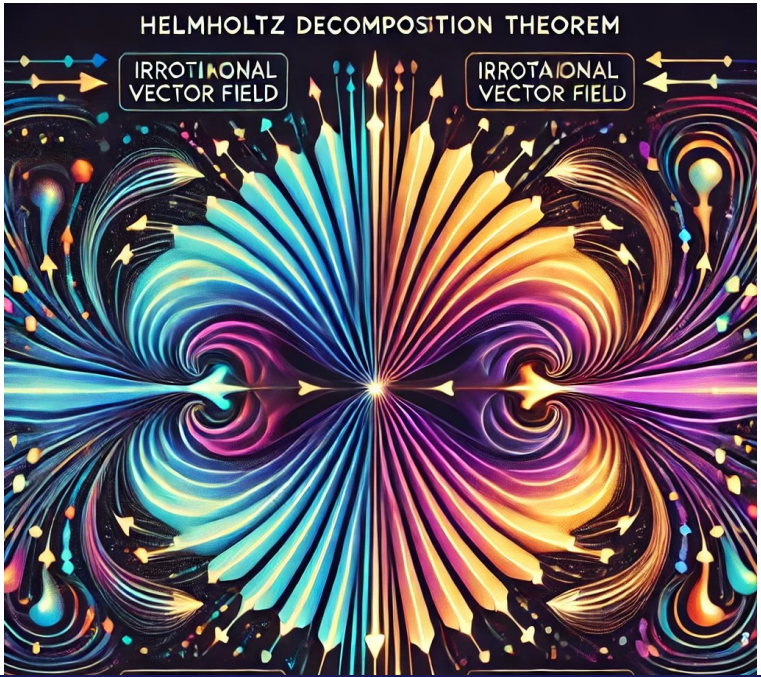
However, the structural approach is:

We study the components of the Helmholz-like **decomposition by Candogan et al. (2011)** of any finite game to a sum of

potential component

harmonic component

non-strategic component



Potential Games

Consider $\mathrm{DirConSet}$ in potential games. Recall

Definition

Potential games are strategic games where exists a function $P : S \to \mathbb{R}$, where for any $s_{-i} \in \times_{j \in N \setminus \{i\}} S_j$ and for any strategies s_i, t_i of player i, $u_i(s_i, s_{-i}) - u_i(t_i, s_{-i}) = P(s_i, s_{-i}) - P(t_i, s_{-i})$.

Monderer and Shapley (1996) proved that the set of finite potential games is isomorphic to the set of congestion games, which is

Definition

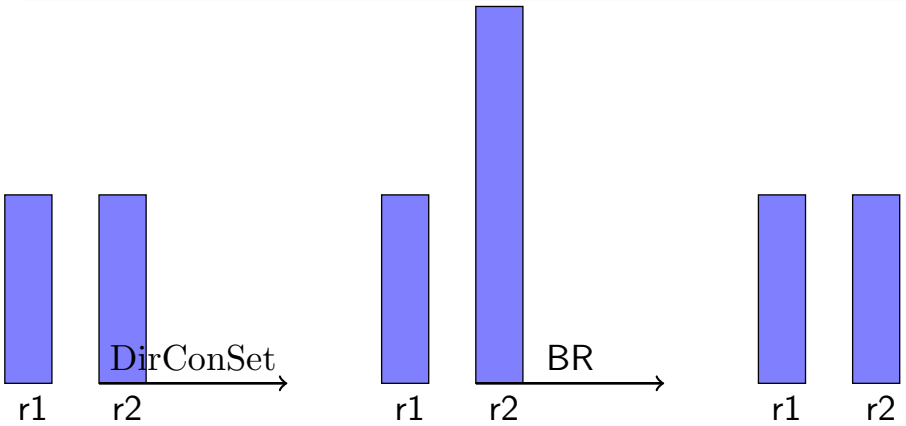
A congestion game consists of $(N, \Sigma, (S_i)_{i \in N}, (c_r)_{r \in \Sigma})$, where each $S_i \subseteq 2^{\Sigma}$.
Each $r \in \Sigma$ has an increasing cost function $c_r : [n] \to \mathbb{R}$, and $C_i(s) \stackrel{\Delta}{=} \sum_{r \in s_i} c_r(l_r(s))$, where $l_r(s) \stackrel{\Delta}{=} |\{j \in N : r \in s_j\}|$.

Potential Games - Overview

$\mathrm{DirConSet}$ in potential games is still $\mathrm{NP\text{-}hard}$, even to approximate within log $\Rightarrow$ we studied

Subclasses

1. Coordination games on graphs are at least as hard as dominating set
2. Singleton congestion games with increasing costs (negative externalities) are solvable polynomially in general position
3. Symmetric congestion games with decreasing costs (positive externalities) are solvable polynomially



Congestion Games - Singleton

Let's zoom in into singleton congestion games with increasing costs.

Definition

A singleton congestion game is a congestion game, where $|s_i| = 1, \forall i \in N, s_i \in S_i$.

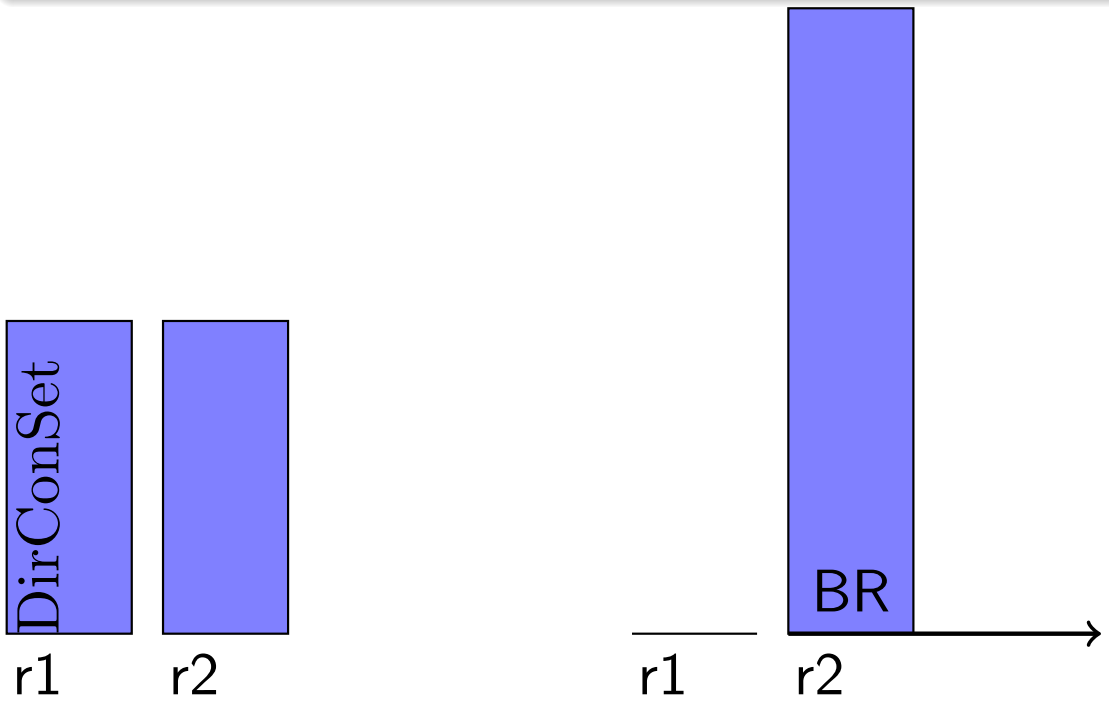
Congestion Games - Singleton

The algorithm will require some definitions.

First, denote $P_r(x) \triangleq \{i \in N : x_i = \{r\}\}$, and for any $A \subseteq R$, let $P_A(x) \triangleq \bigcup_{r \in A} P_r(x)$.

Definition (Intermediate profile, applies to any game)

Given a game $G = (N, (S_i)_{i \in N}, (u_i)_{i \in N})$, profiles $s, d \in S$, and a set of players $E \subseteq N$, denote the intermediate profile $(d_E, s_{N \setminus E})$ by $sd_G(E)$.

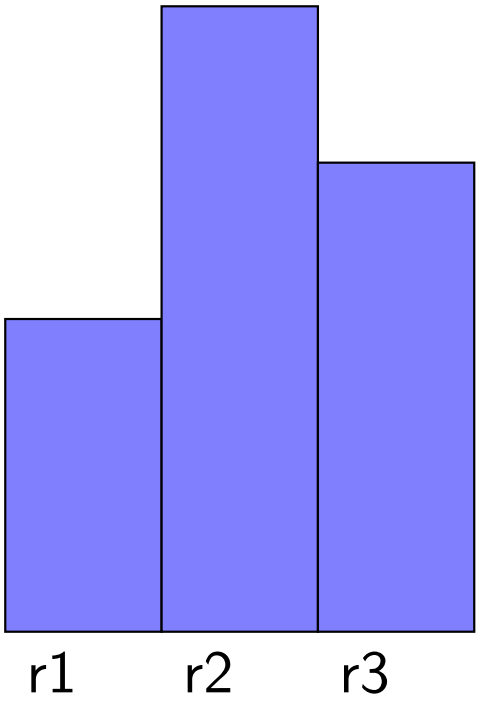


Congestion Games - Singleton - Defs

Definition (Attraction basin, applies to singleton congestion games)

Given a singleton congestion game $(N, R, (S_i)_{i\in N}, (c_j)_{j\in R})$ *and* $s \in S$, *define the attraction basin of* s *in* G *as* $AB_G(s) \stackrel{\Delta}{=} \bigcup_{i\in N}(\cup_{Q\in Br_i(s_{-i})} Q \setminus s_i)$.

Here, for example, if all the costs are $c(x) \stackrel{\Delta}{=} x$, then $AB = \{r1\}$.



Congestion Games - Singleton - Algorithm

Assumption (General position)

For any profile $s \in S$, $|AB(s)| \leq 1$.

Assuming $AB(sd(E)) = \{r\}$, any optimum $\mathrm{DirConSet}(G, s, d)$ is:

$$\{\text{move from } r\} \cup \{\text{move from } p \neq r \text{ to } q \neq r\} \tag{2}$$

$$\cup \{\text{stay at } p \neq r, \text{ while the BR is } \{r\}\} \tag{3}$$

$$\cup \{\text{all } P_r(d) \setminus P_r(s) \text{ OR all but one } P_r(d) \setminus P_r(s) \text{ OR } \emptyset\}. \tag{4}$$

This is a polynomial number of options.
It remains to prove that if more than one of $P_r(d) \setminus P_r(s)$ are missing, then r is the unique BR for everyone.

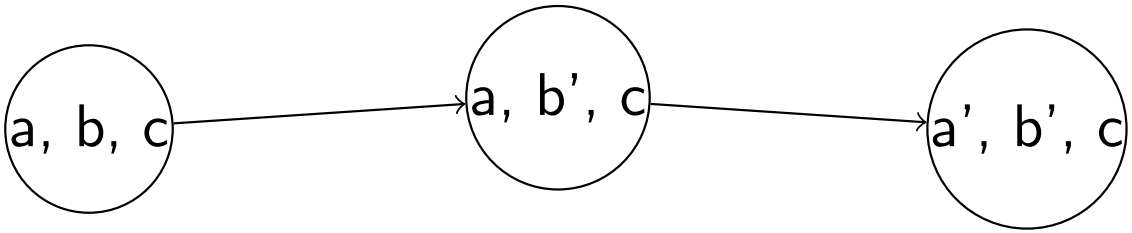
Harmonic Games

To define harmonic games, define the game graph of a finite game:

nodes are strategy profiles,

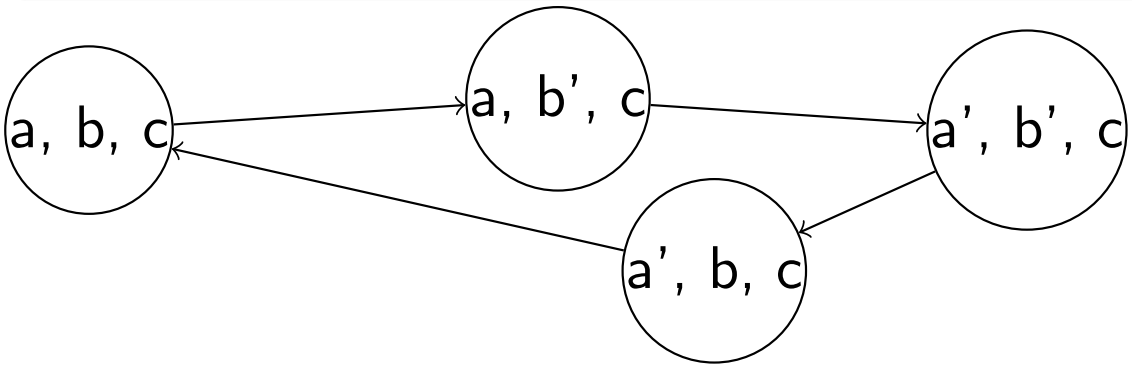
edges mean profiles that differ by exactly one player's strategy.

The utility differences of the unique deviators define a flow.



Definition

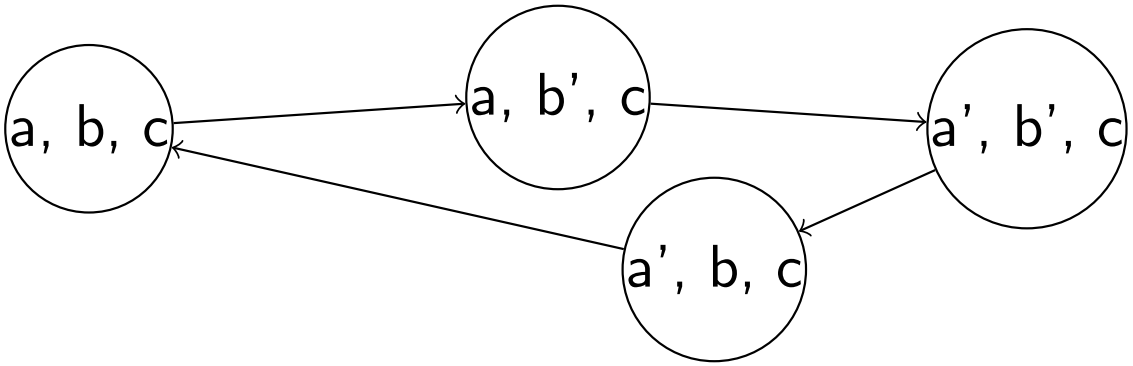
Harmonic games are finite games, which flow representations contain cycles.



Harmonic Games

Definition

Harmonic games *are finite games, which flow representations contain cycles.*



- Cycles indicate a lack of alignment between the players, resembling constant-sum games.
- Indeed, for two-player games, harmonic are equivalent to zero-sum.
- For more players, zero-sum games are a subset of harmonic ones.
- $\Rightarrow$ Study two subclasses: two players OR more players.

Harmonic Games - Two Players

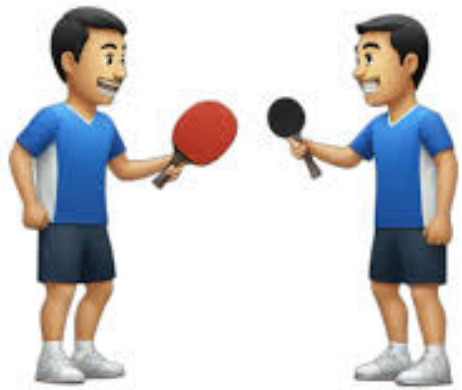
For two players, we study zero-sum games.

Proposition

Two-player zero-sum game has:

if $s, d \in NE$, *then the empty set suffices.*

otherwise, the singleton containing any player suffices, but sometimes so does the empty set.



Harmonic Games - More than Two Players

For more than two players, we prove hardness of zero-sum, implying the hardness of the harmonic games.

Proposition

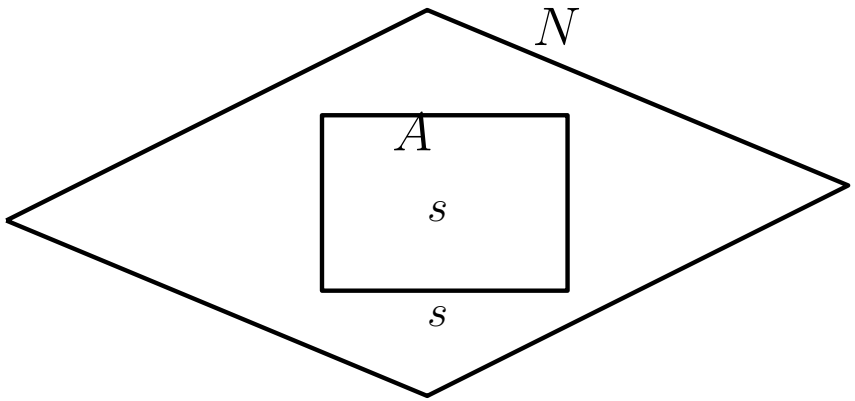
The $\mathrm{DirConSet}$ *in (general) n-player games is at most as hard for approximation, as in* $n+1$*-player zero-sum games.*



Variations of Our Problem

Consider

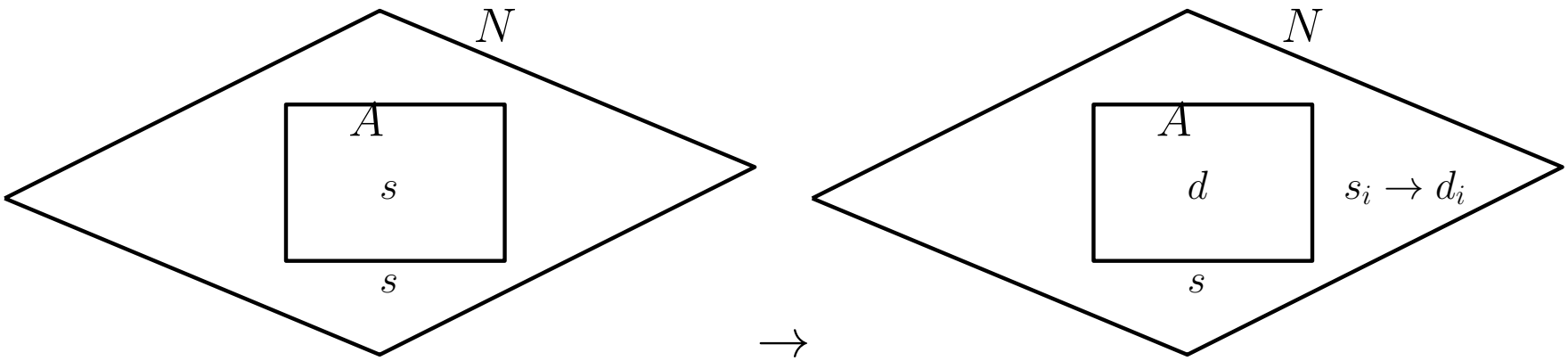
1. Variations on the BR dynamics
2. Variations on the control



Variations of Our Problem

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2. Variations on the control



Order-Independent Direct Control Set

1. Direct control sets motivate the desired **simultaneous** BR
2. Relevant for voting, commitments, and imperfect info
3. Define OI-DirConSet as motivation regardless the order of BR

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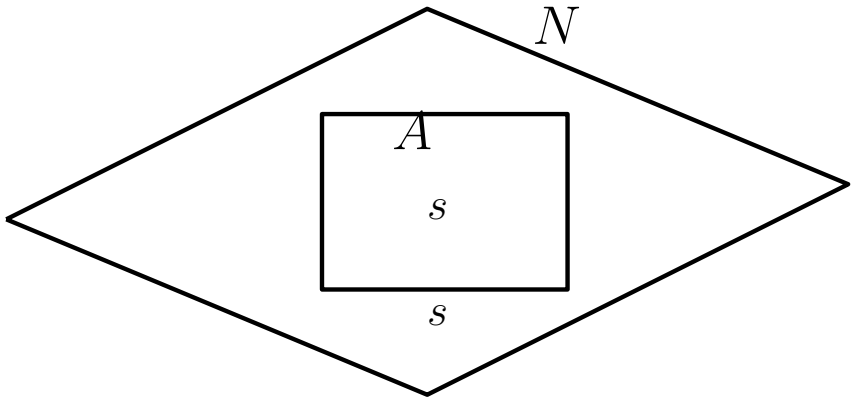
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However,

- Is not only NP-hard (reduction from dominating set),
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$\Rightarrow$ we don't consider OI-DirConSet here



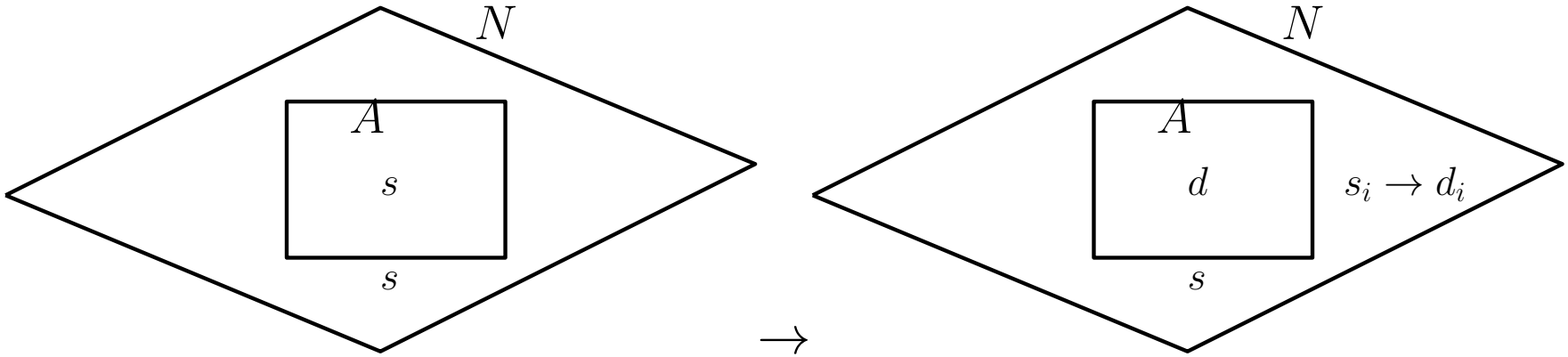
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Sequential Direct Control Set

1. Let's compromise between DirConSet and OI-DirConSet

Sequential Direct Control Set

1. Let's compromise between $\mathrm{DirConSet}$ and $\mathrm{OI\text{-}DirConSet}$
2. by demanding that for a $\mathrm{DirConSet}$ also exists a sequence of best responders.

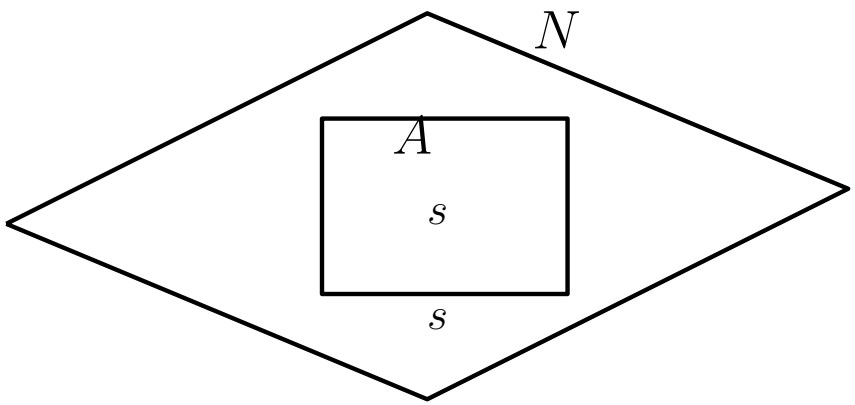
Formally, there exists a sequence of profiles $s(0) \stackrel{\Delta}{=} s, s(1), \ldots, s(T) = d$, such that for any step $t \in \{0, 1, 2, \ldots, T-1\}$, exactly one agent differs between $s(t)$ and $s(t+1)$, and that agent best responds in $s(t+1)$.

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2. This is also hard, even to approximate, but
3. in a singleton congestion game, every $\mathrm{DirConSet}$ is sequential

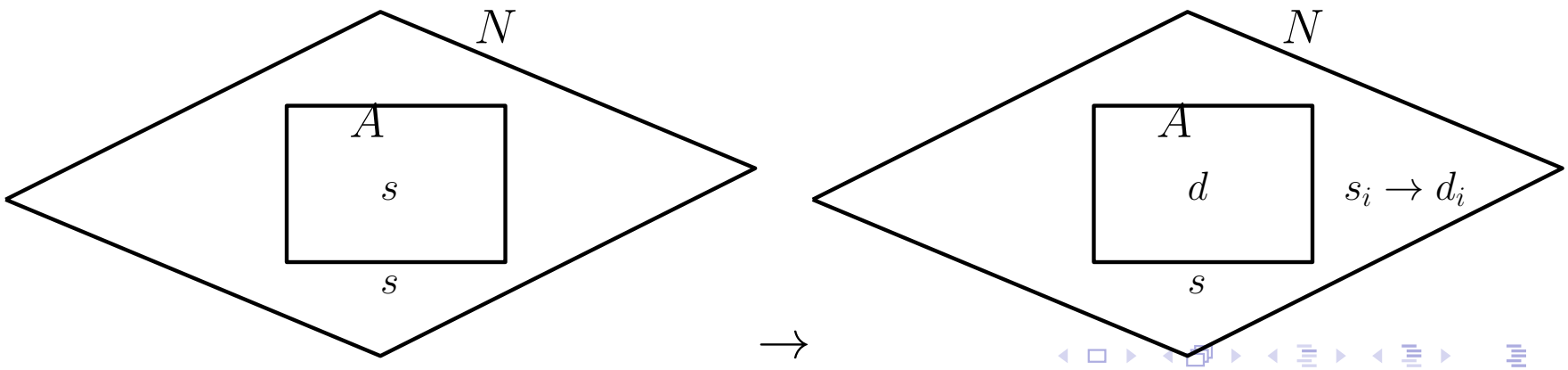


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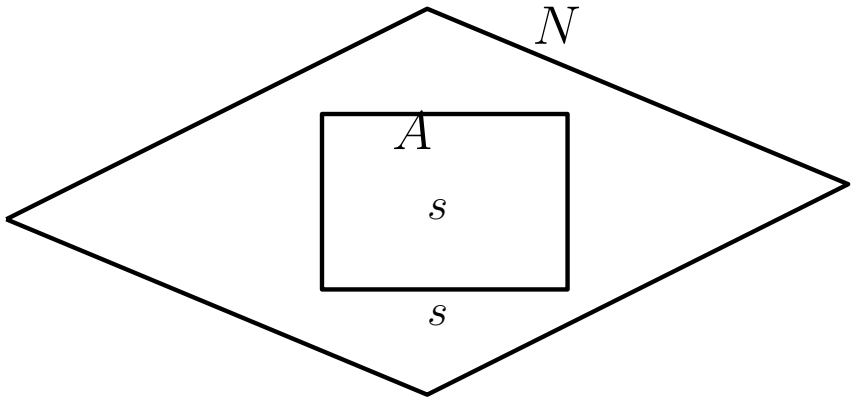
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(Not Necessarily Direct) Control Set

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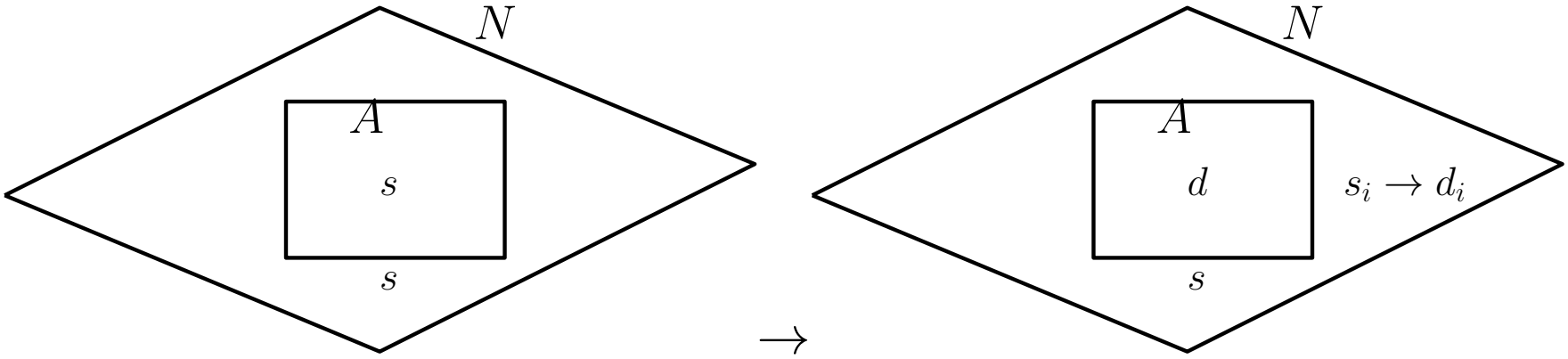
This assumes much more power to the controller!
Sometimes every control set is always direct, e.g. for coordination games



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Conclusions

1. Improving equilibria through agents
2. Always reaches the desired stable set



Future Work

- Sequential $\mathrm{DirConSet}$ and other variations, to motivate better
- Subsidising the control set? Cooperative approaches?
- Characterising equilibria with $\mathrm{DirConSet}$ of certain sizes



Thank You!



Appendix

More slides follow.

Model – Direct Control Set Problem - Representation

Definition (Direct Control Set Problem)

The direct control set problem receives a game $G = (N, S, (u_i)_{i=1,\ldots,n})$, *profile* s *and equilibrium* d, *and weights* $w\colon N \to \mathbb{R}_+$.
A solution is a set of agents $A \subseteq N$.
A feasible solution is a solution that can bring s *to* d *directly,*
Find a feasible solution A *with the minimum* $w(A) \stackrel{\Delta}{=} \sum_{a \in A} w(a)$.

Remark (Representation)

Storing utilities directly is impractical ($\Omega(n \cdot 2^n)$ *space*) $\Rightarrow$ *the utilities are provided by a polynomial oracle.*

Basic Remarks

Remark

When the desired d is weakly dominant, then the empty set suffices.

Remark

Two-player constant-sum game has $\mathrm{DirConNum}(G, s, d) = 0$, *when* $s, d \in NE$. *Indeed, every such profile consists of maxmin strategies, and any maxmin is a best response to any maxmin of the other player.*



Comparison with Stackelberg

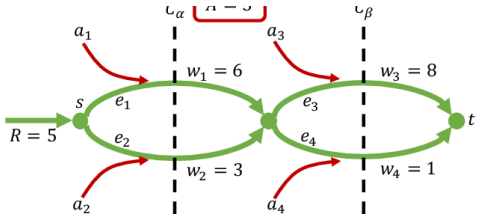
Our model is reminiscent of

Stackelberg strategies (Stackelberg routing in nonatomic routing)

Given a set of players (fraction of flow) to control, control it so that the best responding others maximise the social welfare.

The main differences from our model are

1. It's **dual** to $\mathrm{DirConSet}$: we minimise the set required to directly bring to the desired eq., while Stackelberg strategies maximise the social welfare brought upon using a given set
2. We are **direct**, while Stackelberg strategies are chosen



Optimum Sizes - General Games

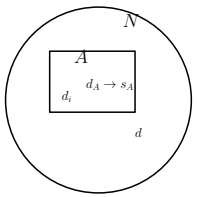
Proposition

For any natural number $n > p \geq 1$, there exists a game G with n players, possessing equilibria s and d, such that $\mathrm{DirConNum}(G, s, d) = p$.

Control vs Strong NE

Proof.

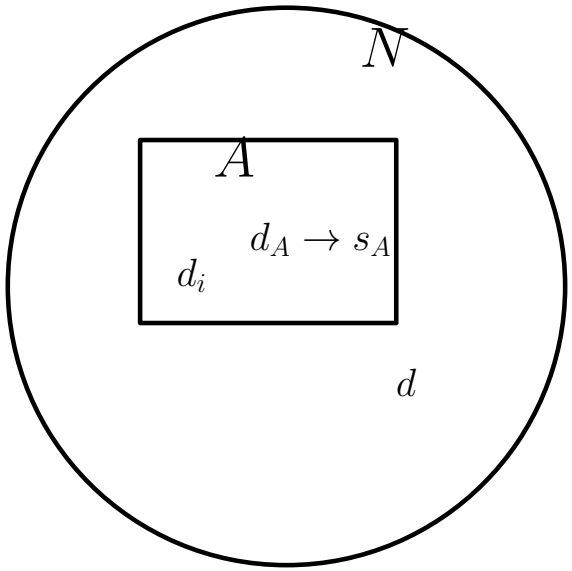
1. Consider a coalitional deviation from d by agents $A \subset N$ to s_A, where $|A| \leq n - m$.
2. By the 1st assumption, the at least $n - (n - m) = m$ non-deviating agents constitute a direct control set with respect to G, s and d.
3. $\Rightarrow \forall i \in A$, playing d_i is a best response now, in particular $u_i(d_{N\setminus A}, s_{A\setminus\{i\}}, d_i) \geq u_i(d_{N\setminus A}, s_{A\setminus\{i\}}, s_i)$.
4. assumption 2 implies that $\forall i \in A$, $u_i(d_{N\setminus\{i\}}, d_i) \geq u_i(d_{N\setminus A}, s_{A\setminus\{i\}}, d_i) \geq u_i(d_{N\setminus A}, s_{A\setminus\{i\}}, s_i)$. Namely, no deviator enjoys the colational deviation $\Rightarrow d$ is $n - m$ strong NE.



Control vs Strong NE

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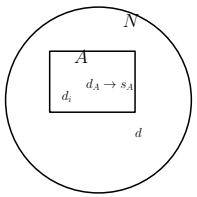
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2. The non-deviating agents constitute a direct control set
3. $\Rightarrow \forall i \in A$, playing d_i is a best response now
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Control vs Strong NE

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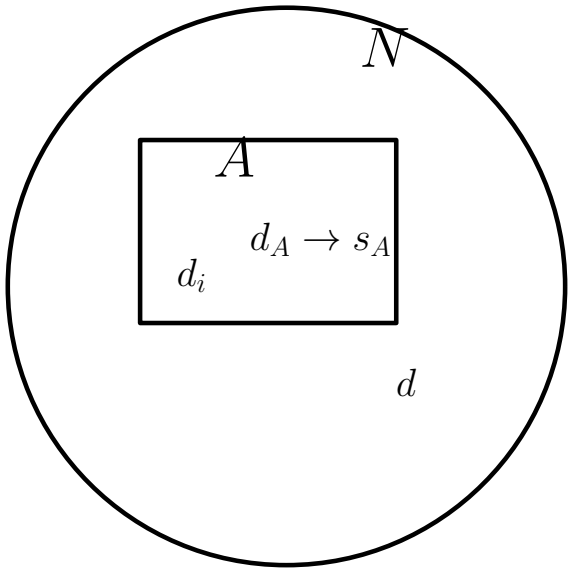
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Hardness - Decision

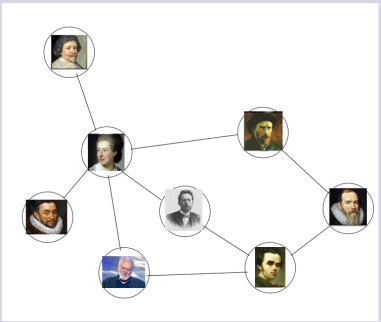
Theorem

The decision version of direct control set is NP-complete.

Proof.

NP-hardness holds even for coordination games on graphs, where

1. Players are nodes;
2. A player's strategy is its chosen colour;
3. A player's utility is the number of her neighbours choosing the same colour.



These problems belong to NP, the witness being a control set (poly). □

Hardness to Approximation

Theorem

Is not approximable within factor $n^{1-\epsilon}$, $\forall \epsilon > 0$, *unless* $P = NP$.
Holds even when bringing an NE to an NE, and each player has 2 strategies.

Proof.

Reduction from the problem of deleting a minimum vertex set from a connected graph $G = (V, E)$ to obtain a tree. □

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We will cope by dealing with the components of the Helmholz-like decomposition by Candogan et al. (2011) of any finite game to a sum of

potential component

harmonic component

non-strategic component

Monotonicity-Based Algorithms - Monotonicity

Definition

Monotonicity w.r.t. inclusion *means that if A is a* $\mathrm{DirConSet}$ *w.r.t.* (G, s, d)*, then any including set* $A' \supset A$ *is also a* $\mathrm{DirConSet}$ *w.r.t.* (G, s, d)*.*

Monotonicity does not generally hold, even in singleton congestion games, but we'll assume it by default.

We'll even assume monotonicity w.r.t. motivating each single player.

Monotonicity-Based Algorithms - Player-wise Monotonicity

Consider the agents required to control a given agent's best response.

Definition

Given agent $k \in N$, profile s and Nash eq. d, if either $k \in A$, or the agents $A \subseteq (N \setminus \{k\})$ can play $(d_i)_{i \in A}$ such that agent k's best response to $(d_i)_{i \in A}, (s_j)_{j \in N \setminus A}$ is d_k, then we say A can (directly) bring k from s to d. Such a set A is called a direct control set for k with respect to G, s, and d.

Observation

Set $A \subseteq N$ is a direct control set with respect to G, s and d $\iff$ A is a direct control set for k with respect to G, s and d, for each $k \in N \setminus A$.

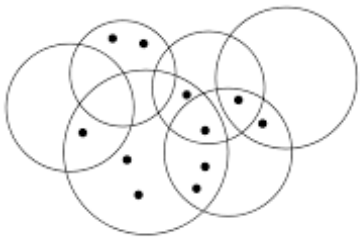
We are going to assume the following stronger monotonicity

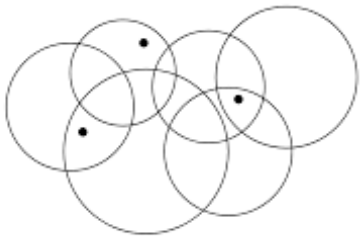
Definition

Player-wise monotonicity with respect to inclusion means that $\forall k \in N$: if $A \in \mathrm{DirConSet}_k(G, s, d) \Rightarrow$ so is any including set $A' \supset A$.

Assume that **for each player**, every minimal direct control set is a singleton.
For approximation, given a direct control set problem (G, s, d), reduce it to hitting set as follows:

1. For each $i \in N$, check which singletons constitute direct control sets for that player. Denote their union by $DCS(i)$. ($i \in DCS(i)$.)
2. Return the hitting set for the collection $(DCS(i))_{i \in N}$.


A hitting set problem


A minimal hitting set

Monotonicity-Based Algorithms - Reductions - Minimum Hitting Set of Bundles

Another interesting case occurs when for each player, we are given all the minimal direct control sets. Can be dealt with using the Minimum Hitting Set of Bundles.

Monotonicity-Based Algorithms - Local Ratio (LR)

The idea: local ratio is extended inductively

$C \leftarrow \emptyset$

While $\exists$ a player $i \in N \setminus C$, s.t. $d_i \notin BR_i(d_C, s_{N \setminus C})$

1. Let $N[i] \leftarrow \{i\} \cup \{j \in N \setminus C : s_j \text{ can influence } u_i\}$
2. Set $\epsilon \leftarrow \min_{j \in N[i]} w(j)$
3. $w'(i) \leftarrow w(i)$
4. $w(j) \stackrel{\Delta}{=} w(j) - \begin{cases} \epsilon & \text{if } j \in N[i], \\ 0 & \text{otherwise} \end{cases}$
5. if $\forall j \in N[i], w(j) = 0$ then $\Rightarrow w(i) \leftarrow w'(i)$
6. $C \leftarrow C \cup \{k \in N : w(k) = 0\}$

Since the optimum picks at least one and at most $|N[i]| - 1$, the local ratio of the picked solution to the optimum is $\leq$ the maximum intersection number.

$\Rightarrow$ also the total ratio.

Congestion Games - Optimum Sizes

First, recall that

Proposition

For any natural numbers $n > p \geq 1$, there exists a game G with n players, possessing equilibria s and d, such that $\mathrm{DirConNum}(G, s, d) = p$.

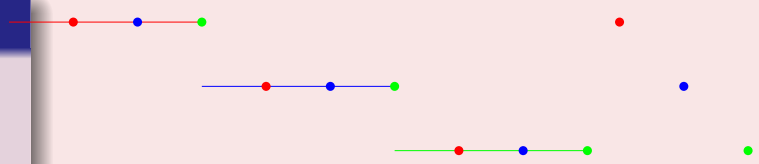
Interestingly, for congestion games, the upper bound does not decrease

Proposition

For any $n > p \geq \frac{n-1}{2}$, there exists a congestion game G with n players, possessing equilibria s and d, such that $\mathrm{DirConNum}(G, s, d) = p$.

Proposition

For any $n > p \geq \frac{n-1}{2}$, there is a congestion game with $\mathrm{DirConNum}(G, s, d) = p$.



Proof.

For any n, let $N \triangleq [n]$ and $\Sigma \triangleq \left[n^2 + n\right]$, and define the costs: For any $e \in \left[n^2\right]$, let

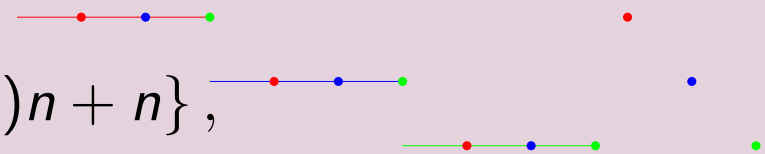
$$c_e(x) \triangleq \begin{cases} x & x \geq 2, \\ 0 & \text{otherwise.} \end{cases}$$

For any $e \in \left\{n^2 + 1, \ldots, n^2 + n\right\}$, let (for a fixed p)

$$c_e(x) \triangleq \begin{cases} 2(2p + 1 - n) \cdot x & x \geq 1, \\ 0 & \text{otherwise.} \end{cases}$$

Proof.

Now, let player i's strategy set be



$$S_i \triangleq \{\{(i-1)n+1, (i-1)n+2, \ldots, (i-1)n+n\},$$
$$\{i, n+i, 2n+i, \ldots, (n-1)n+i\} \cup \{n^2+i\}\}.$$

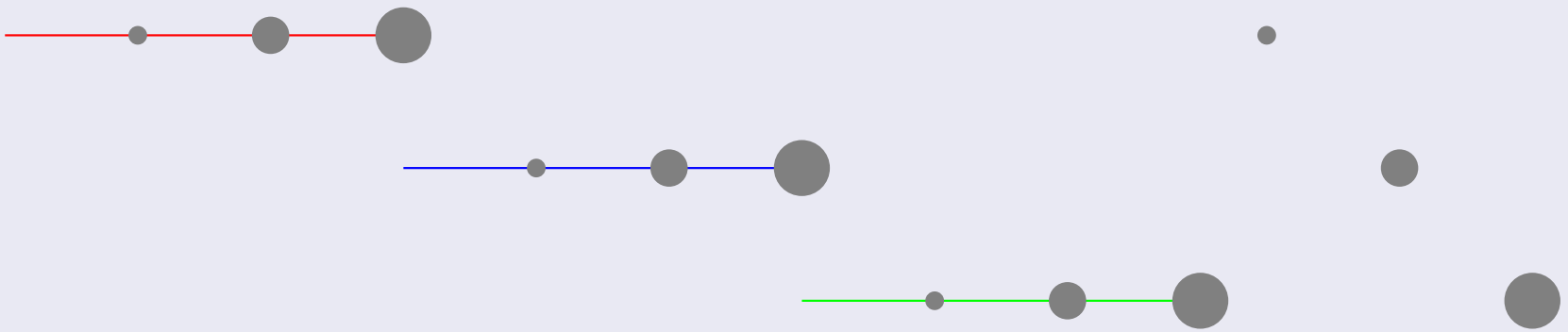
Everyone playing the first strategy is an NE (no intersection), and we now prove that everyone playing the second one is an NE, too. Moving from the first NE to the second one requires at least p players to move. Indeed, when q players play the dots-strategies, keeping playing the lines-strategy costs $q \cdot 2$, while deviating to the dots-strategy would incur $(n-q-1)\cdot 2 + 2(2p+1-n) = 2(2p-q)$, and $q \cdot 2 < 2(2p-q) \iff q < p$.

Substituting $q \leftarrow n-1$, implies everyone playing the second strategy is an NE. □

Proof Illustration for $n = 3$.

1. Player 1 either plays the red line (i.e. dots $1, 2, 3$), or all the small "$\cdot$" dots (that is, dots $1, 4, 7, 10$).
2. Player 2 either plays the blue line (dots $4, 5, 6$), or all the average "$\bullet$" dots (dots $2, 5, 8, 11$).
3. Player 3 either plays the green line (dots $7, 8, 9$), or all the big "$\bullet$" dots (dots $3, 6, 9, 12$).

$\Rightarrow$ To motivate i to move from the "line"-NE to the "dots"-NE, at least p (depending on the costs) other players have to move first.



Congestion Games - Singleton

Let's zoom in into singleton congestion games with increasing costs.

Definition

A singleton congestion game is a congestion game, where $|s_i| = 1, \forall i \in N, s_i \in S_i$.

Theorem

For singleton congestion games with n players and m resources, there always exists a $\mathrm{DirConSet}$ *of size at most* $n - \lceil n/m \rceil$*, and this is tight.*

Proof.

When bringing s to d, consider a most loaded resource in d. Then, all the players who play any other resource in d form a $\mathrm{DirConSet}$ of size at most $n - \lceil n/m \rceil$.

For tightness, intuitively, consider an example, where the motivated movement after the controlled players have moved is towards just one resource. Therefore, $s_i \neq d_i, \forall i \in N \Rightarrow \mathrm{DirConNum} \geq n - \lceil n/m \rceil$. □

Congestion Games - Singleton

Theorem

For singleton congestion games with n players and m resources, there always exists a $\mathrm{DirConSet}$ *of size at most $n - \lceil n/m \rceil$, and this is tight.*

A formal tight example.

Proof.

Consider a singleton congestion game G with $c_j(x) \stackrel{\Delta}{=} m(x-1)+j$. Let s and d fulfil $s_i \neq d_i, \forall i \in N$, and $\lfloor n/m \rfloor \leq |P_r(d)| \leq \lceil n/m \rceil$, for any $r \in R$. In any profile, no two resources have equal costs $\Rightarrow |AB(sd(E))| \leq 1$, for any $E \subseteq N$. All the players in $N \setminus P_{AB(sd(E))}(d)$ need to move, therefore they all have to belong to the $\mathrm{DirConSet}$
$\Rightarrow |\mathrm{DirConSet}| \geq n - \lceil n/m \rceil$. □